iCyPhy

Semantic Service Discovery for The Connected Car with Accessors

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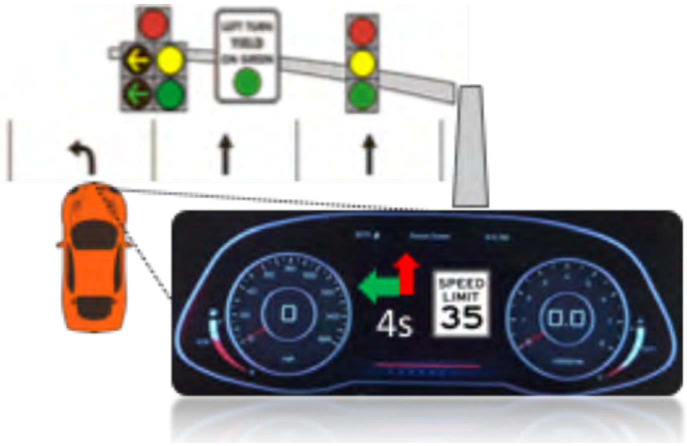
Connected Cars

- Connected cars have the potential to transform a vehicle from a transportation platform to a platform for integrating humans with a city.
- Communicate with services:
  – Vehicle to vehicle
  – Vehicle to infrastructure
  – Vehicle to web service

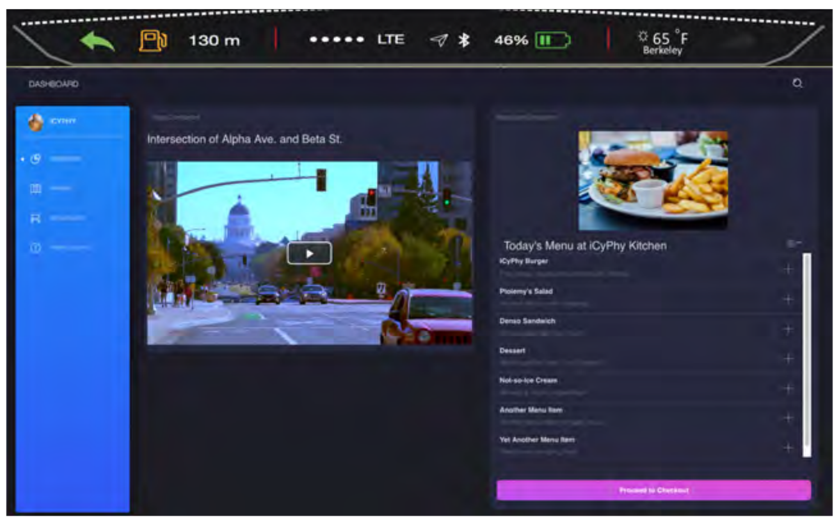


Challenges

- Once a driver has established the intention of seeking out a service (eg. parking), how does the connected vehicle:
  - Select the most contextually appropriate service from a service library?
  - Communicate with this service, with which it has never before interacted, using the correct radio, protocol, and API?
  - Coordinate its use of the service with on-board sensors or other similarly discovered services?

Overview

- Ontologies
- Semantic Accessors
  - i.e. local proxies for remote services
- The Semantic Accessor Architecture
- Demo: Dynamic Dashboard User Interface



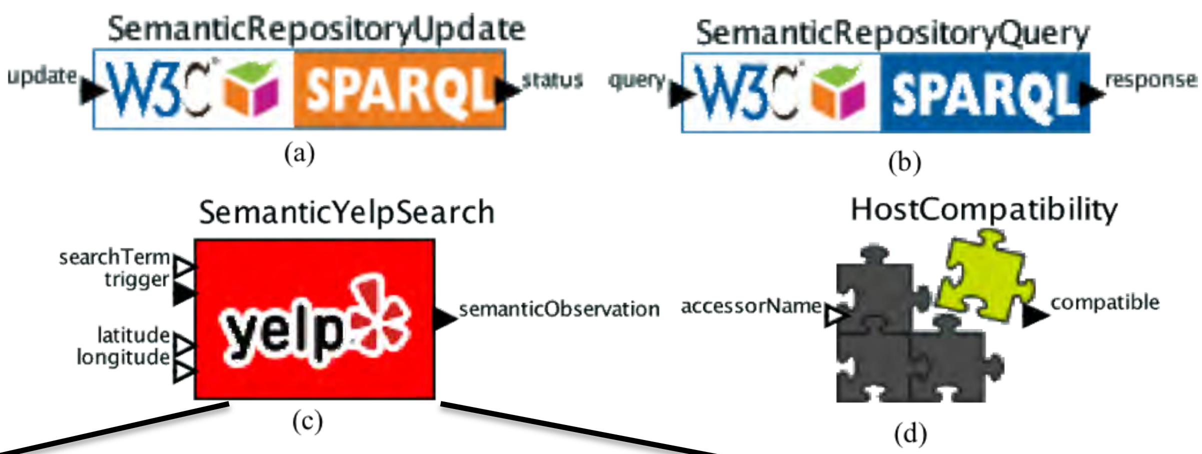
Background: The Semantic Web Stack

- Resource Description Framework (RDF)
  - An abstract model for expressing relationships between concepts
  - Expressed as subject, predicate, object triples
    - "A cow” (subject) "is a" (predicate) "farm animal" (object)
    - "The mall parking lot" (subject) "has the number of free spaces” (predicate) "45" (object)

- Semantic Repository
  - A Triple Store database
  - Accessible via the SPARQL Protocol and RDF Query Language (SPARQL)

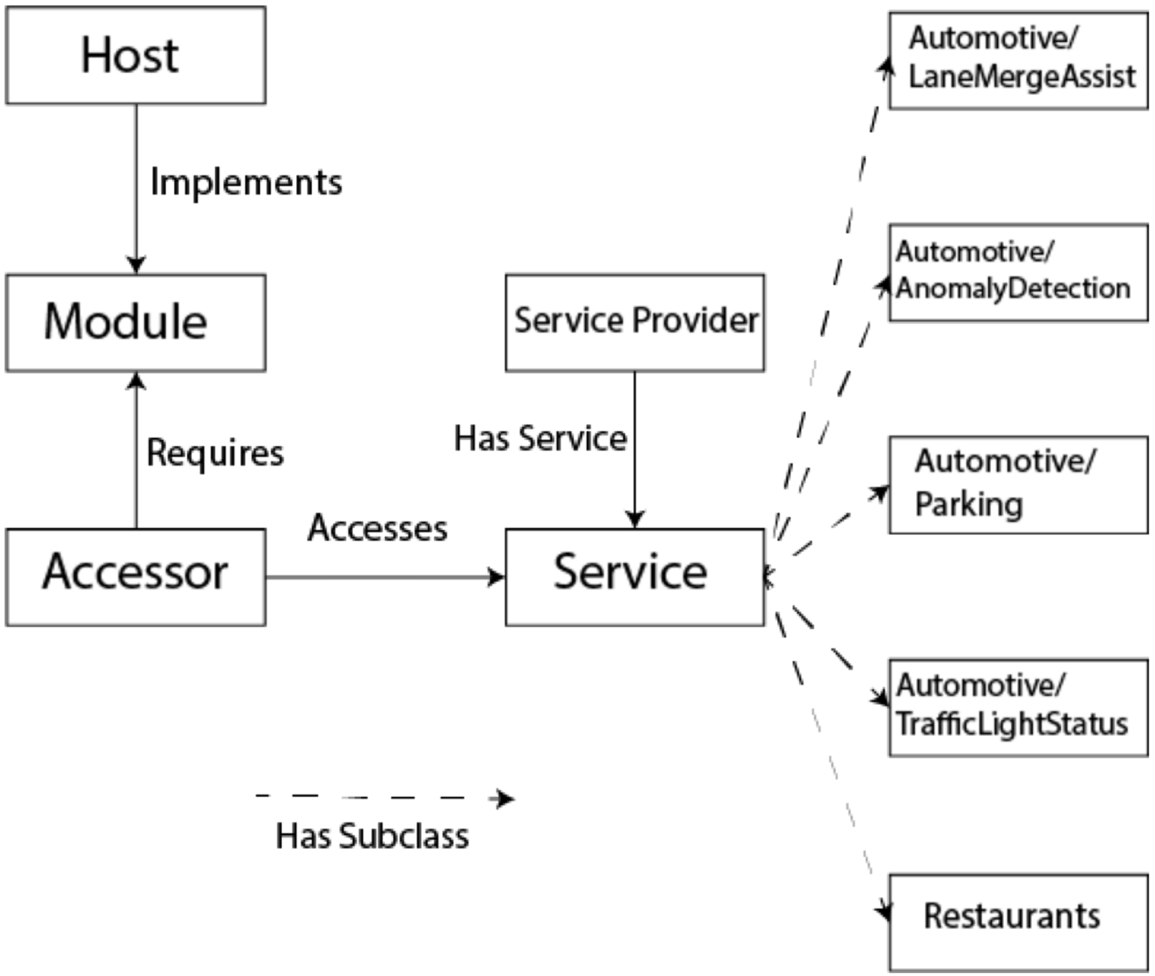
Semantic Accessors

- An accessor is a downloadable chunk of JavaScript implementing a local proxy for a remote service.
- A *semantic* accessor interacts with semantic technologies

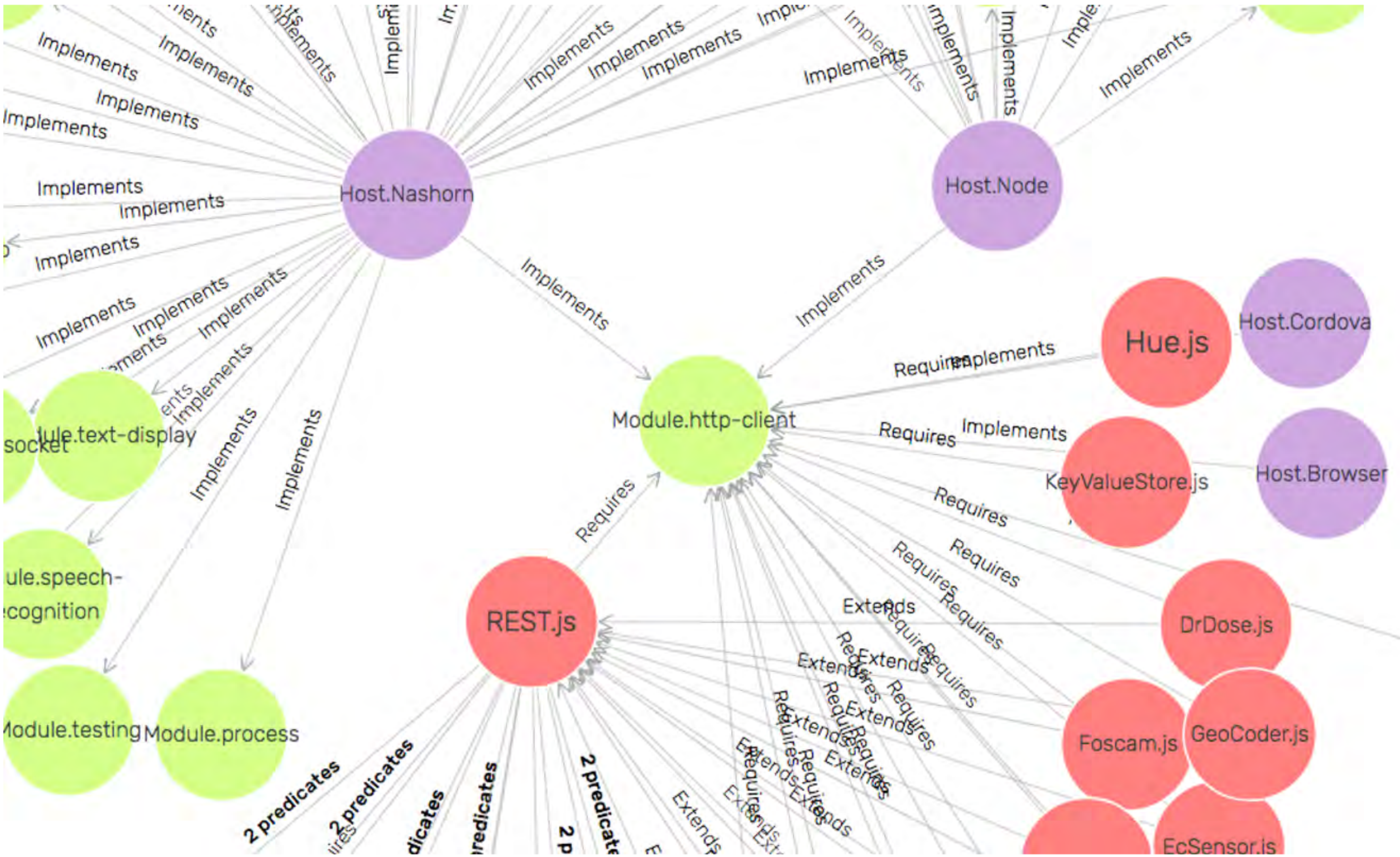



```
exports.handleResponse = function(message){
  var writer = N3.Writer({ prefixes:
   { sosa: 'http://www.w3.org/ns/sosa/',
     rdf: 'http://www.w3.org/1999/02/22-rdf-syntax-
         ns#',
     xsd: 'http://www.w3.org/2001/XMLSchema#',
     schema: 'http://schema.org/'
    }});
  var obsNode = store.createBlankNode('
      SemanticYelpSearchObservation');
    ...
}
```

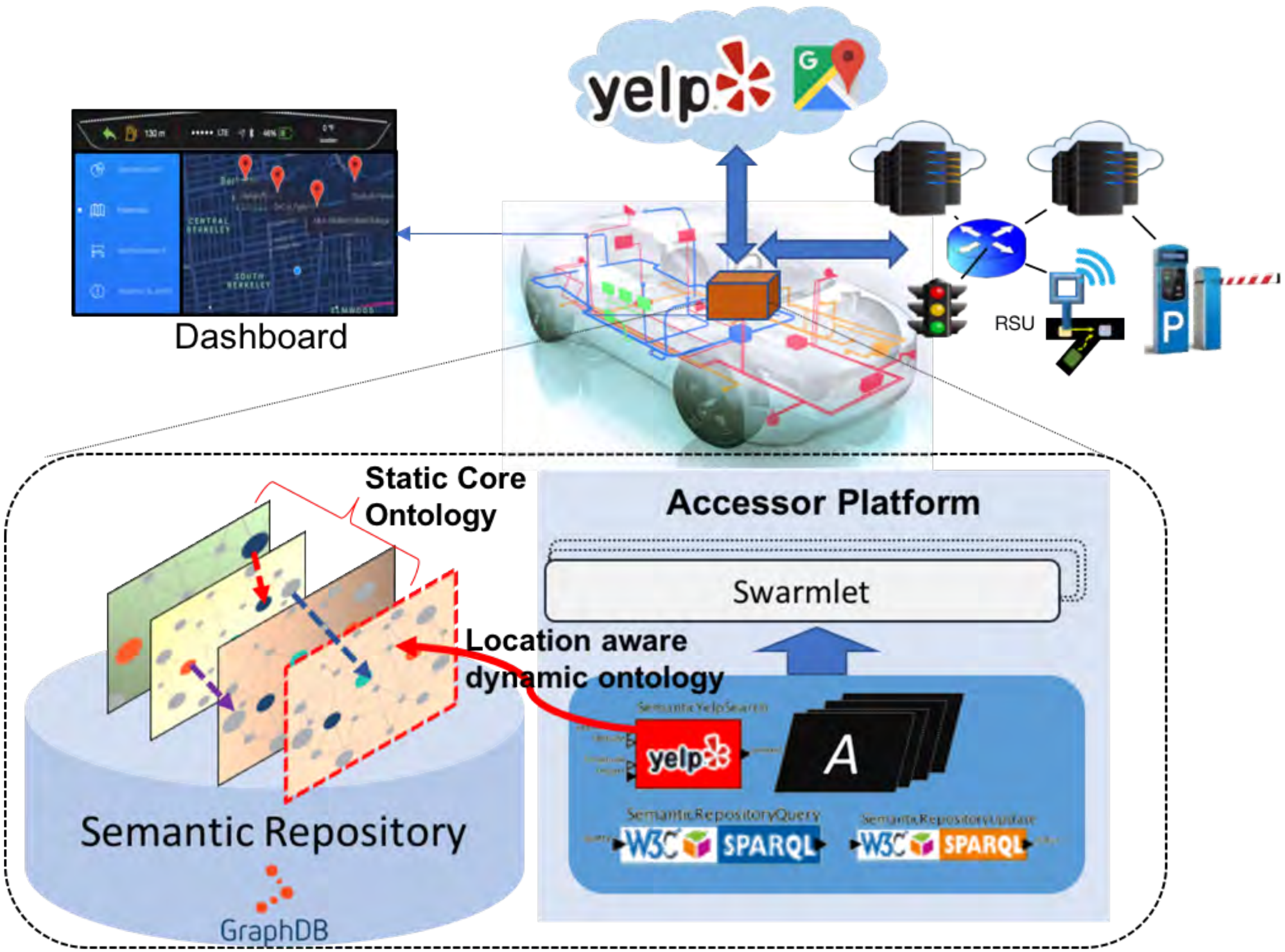
Accessors and Services Ontology



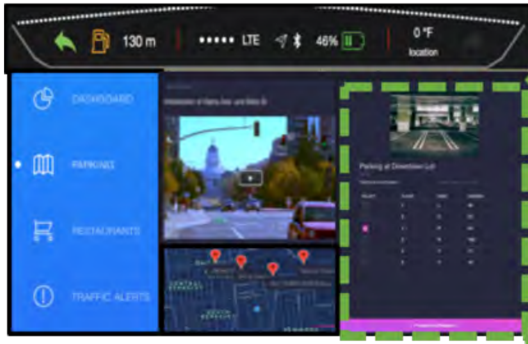
Accessors Ontology Snapshot



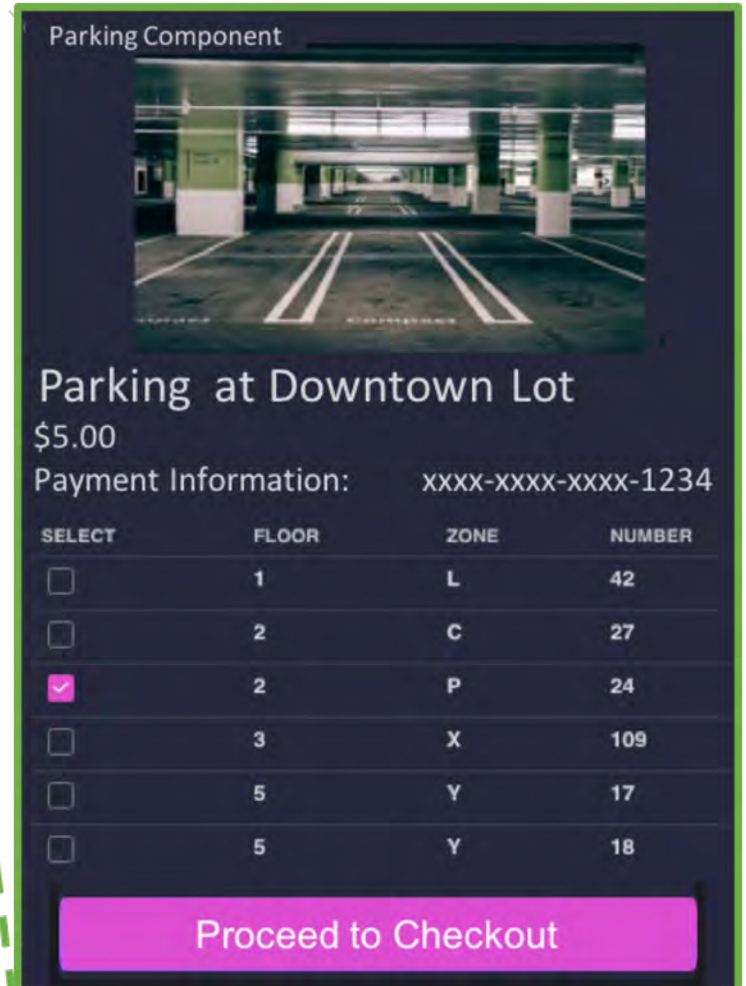
The Semantic Accessor Architecture



Web Components Enable a Dynamic UI

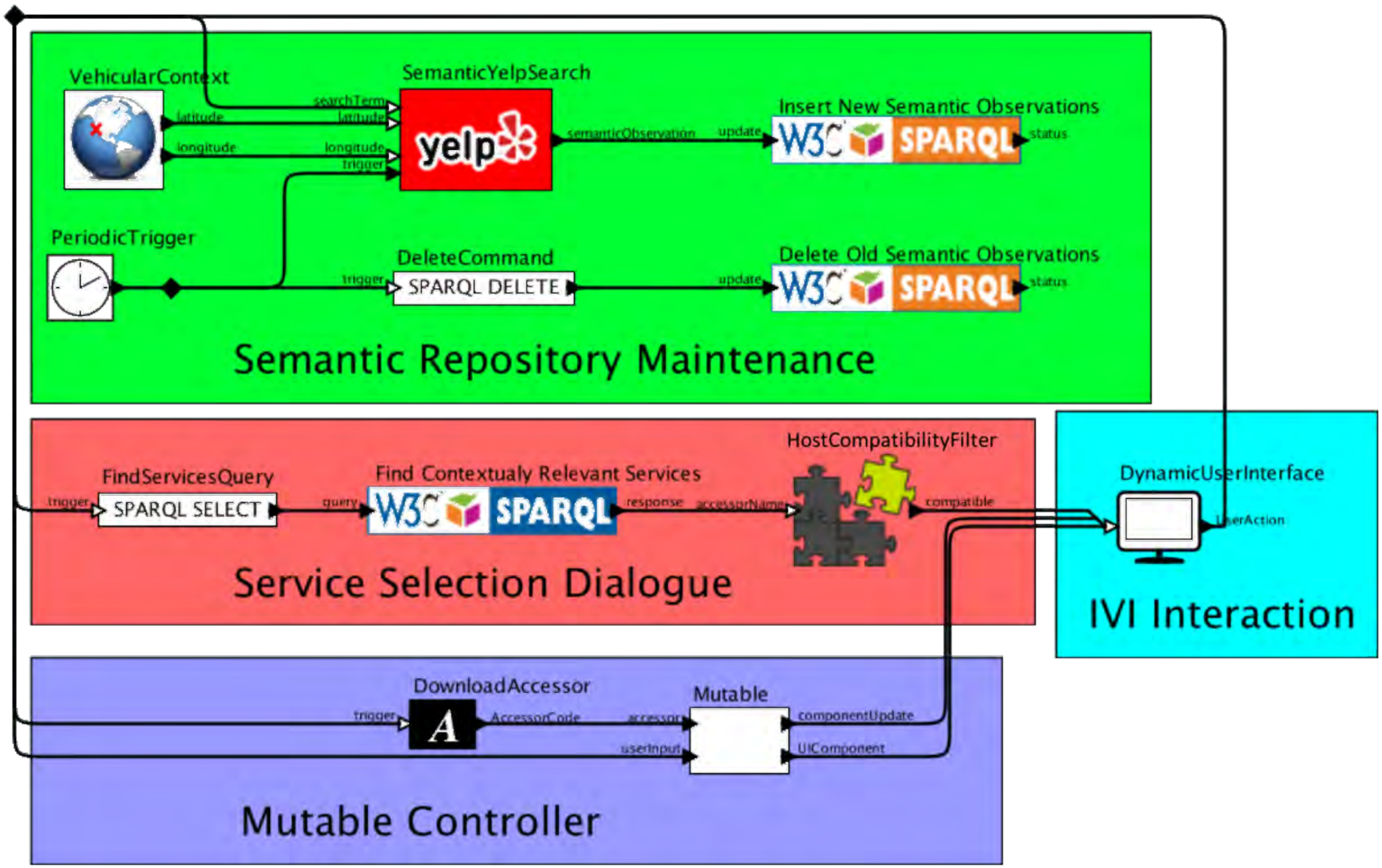


Dashboard



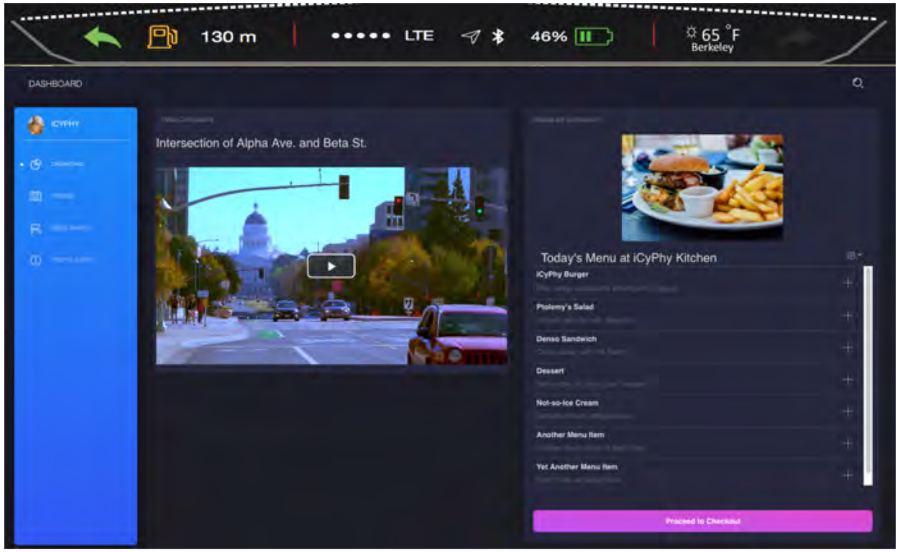
The downloaded service produces its own user interface.

Swarmlet Controller



User Interface Demo

- Intended as a proof of concept
- Significant human-factors engineering needed for a real vehicle



Conclusion

- We addressed challenges of semantic service discovery for connected cars
  - Accessor Ontology
  - Semantic Accessors
  - Semantic Accessor Architecture
  - Dynamic User Interface

Future Work

- Use inference to enhance the knowledge stored in the semantic repository
- Dynamically discover semantic sensors
  - a semantic repository might accumulate data sources as it progressively infers newly relevant sensors.
- Improve security mechanisms
  - ”locally centralized, globally distributed” authentication (Kim and Lee, 2017)

[1] H. Kim and E. A. Lee, “Authentication and Authorization for the Internet of Things,” *IT Professional*, vol. 19, no. 5, pp. 27–33, 2017.

Questions